HDOC Day 32

Aim

To write a menu-driven C program using separate functions and variable-length two-dimensional arrays to verify that matrix addition is commutative and demonstrate that matrix multiplication is not commutative in general.

Theory

Matrix addition is commutative: A + B = B + A for matrices of the same dimensions. Matrix multiplication is generally not commutative: AB may differ from BA. For the multiplication comparison, square matrices of the same order are used so both products are defined and comparable. Some special matrices can have AB = BA.

Algorithm

1. Start. Display a menu with addition, multiplication and exit options.

2. Addition: Read rows and columns, then both matrices. Using a separate function, calculate A+B and B+A, compare corresponding elements and display both matrices and the result.

3. Multiplication: Read the order of two square matrices and their elements. Using a separate function, calculate AB and BA with three nested loops, compare corresponding elements and display both products.

4. Repeat until the user selects Exit; stop.

C Program

```
#include <stdio.h>

void read(int r, int c, int a[r][c]) {
    for (int i = 0; i < r; i++)
        for (int j = 0; j < c; j++)
            scanf("%d", &a[i][j]);
}
void print(int r, int c, int a[r][c]) {
    for (int i = 0; i < r; i++) {
        for (int j = 0; j < c; j++)
            printf("%d ", a[i][j]);
        printf("\n");
    }
}
void addition(int r, int c, int a[r][c], int b[r][c]) {
    int ab[r][c], ba[r][c], same = 1;
    for (int i = 0; i < r; i++)
        for (int j = 0; j < c; j++) {
            ab[i][j] = a[i][j] + b[i][j];
            ba[i][j] = b[i][j] + a[i][j];
            if (ab[i][j] != ba[i][j]) same = 0;
        }
    printf("A + B:\n"); print(r, c, ab);
    printf("B + A:\n"); print(r, c, ba);
    printf("Addition commutative: %s\n", same ? "Yes" : "No");
}
void multiply(int n, int a[n][n], int b[n][n]) {
    int ab[n][n], ba[n][n], same = 1;
    for (int i = 0; i < n; i++)
```

```
        for (int j = 0; j < n; j++) {
            ab[i][j] = ba[i][j] = 0;
            for (int k = 0; k < n; k++) {
                ab[i][j] += a[i][k] * b[k][j];
                ba[i][j] += b[i][k] * a[k][j];
            }
            if (ab[i][j] != ba[i][j]) same = 0;
        }
    printf("A x B:\n"); print(n, n, ab);
    printf("B x A:\n"); print(n, n, ba);
    printf("Products %s.\n", same ? "are equal for these matrices" : "are different (not commutative)");
}
int main(void) {
    int choice, r, c;
    while (1) {
        printf("\n1. Addition  2. Multiplication  3. Exit\n");
        printf("Enter choice: ");
        if (scanf("%d", &choice) != 1) return 0;
        if (choice == 3) break;
        if (choice == 1) {
            printf("Enter rows and columns: ");
            scanf("%d%d", &r, &c);
            if (r <= 0 || c <= 0) { printf("Invalid size\n"); continue; }
            int a[r][c], b[r][c];
            printf("Enter matrix A:\n"); read(r, c, a);
            printf("Enter matrix B:\n"); read(r, c, b);
            addition(r, c, a, b);
        } else if (choice == 2) {
            printf("Enter order of square matrices: ");
            scanf("%d", &r);
            if (r <= 0) { printf("Invalid size\n"); continue; }
            int a[r][r], b[r][r];
            printf("Enter matrix A:\n"); read(r, r, a);
            printf("Enter matrix B:\n"); read(r, r, b);
            multiply(r, a, b);
        } else printf("Invalid choice\n");
    }
    return 0;
}
```

Sample Output

```
1. Addition  2. Multiplication  3. Exit
Enter choice: 1
Enter rows and columns: 2 2
Enter matrix A:
1 2
3 4
Enter matrix B:
5 6
7 8
A + B:
6 8
10 12
B + A:
6 8
10 12
Addition commutative: Yes

1. Addition  2. Multiplication  3. Exit
Enter choice: 2
Enter order of square matrices: 2
Enter matrix A:
1 2
3 4
Enter matrix B:
5 6
```

```
7 8
A x B:
19 22
43 50
B x A:
23 34
31 46
Products are different (not commutative).

1. Addition  2. Multiplication  3. Exit
Enter choice: 3
```

Result

The menu-driven C program was executed successfully. It verified that matrix addition is commutative and demonstrated, using a counterexample, that matrix multiplication is not commutative in general.